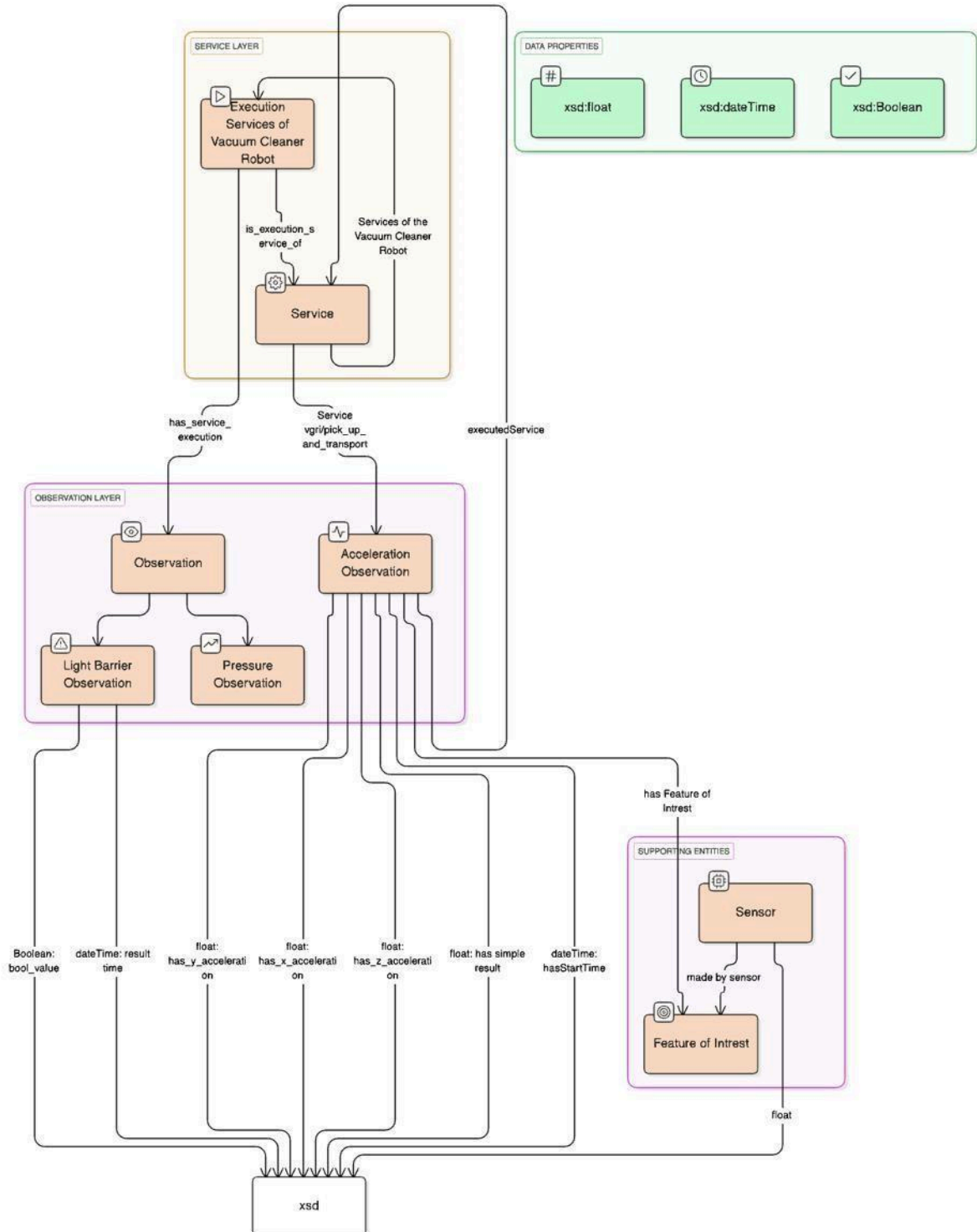


Exercise 6

Visual Sketch:



Summary:

This visual model represents how five executions of the Vacuum Gripper Robot (VGR_1) service are semantically integrated into the ontology using the SOSA ontology for

observations and sensor data. The model is layered to separate service execution, sensor readings, and supporting components.

Main Components:

Service Execution Layer:

- **Service Class** represents general manufacturing actions. A specific service, `vgr/pick_up_and_transport`, is the focus of this modelling.
- **Execution Services of Vacuum Gripper Robot** captures five actual executions of the service.
- Each service execution instance is linked to the service via:
 - `executesService` (points to the abstract service that was executed);
 - `is_execution_service_of` (links the instance to the general `VGR_1` execution class).

Each execution instance is annotated with `hasStartTime` and `hasEndTime` using `xsd:dateTime` for precise tracking allowing for temporal reasoning (execution order, duration).

Observation Layer (SOSA-based)

Each execution is associated with three distinct types of observations, modelled as **subclasses of Sosa: Observation**.

a) Pressure Observation captures the pressure value using `has_simple_result` with `xsd:float` and linked to a **Sensor** (via `made by sensor`) and a **Feature of Interest**.

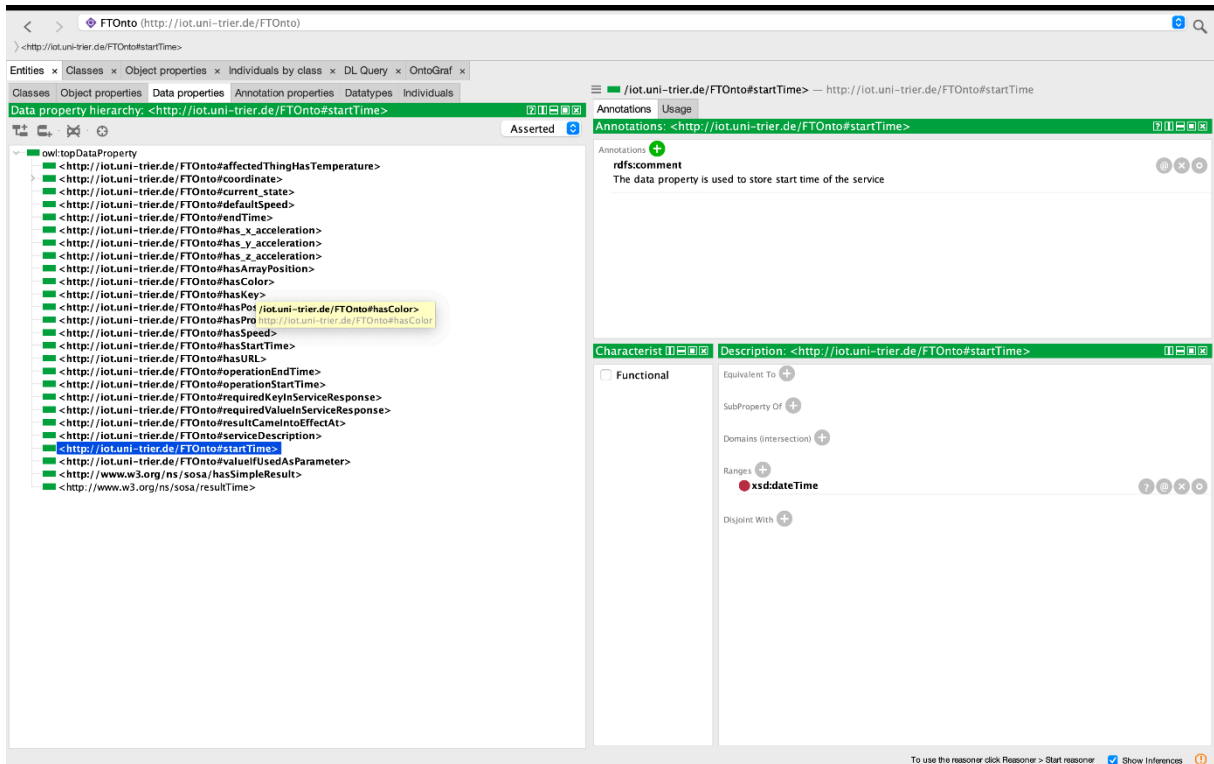
b) Acceleration Observation tracks motion along 3 axes: (`has_x_acceleration`, `has_y_acceleration`, `has_z_acceleration`). These are `xsd:float` data properties, linked to a **Sensor** mounted on the VGR's crane and a **Feature of Interest** like the crane's movement. Time of observation is stored in `resultTime`.

c) Light Barrier Observation records whether the light barrier at the waiting platform was interrupted, uses a Boolean data property: `bool_value` of type `xsd:Boolean`, connected to its sensor and feature of interest and also uses `resultTime` to track when the detection occurred.

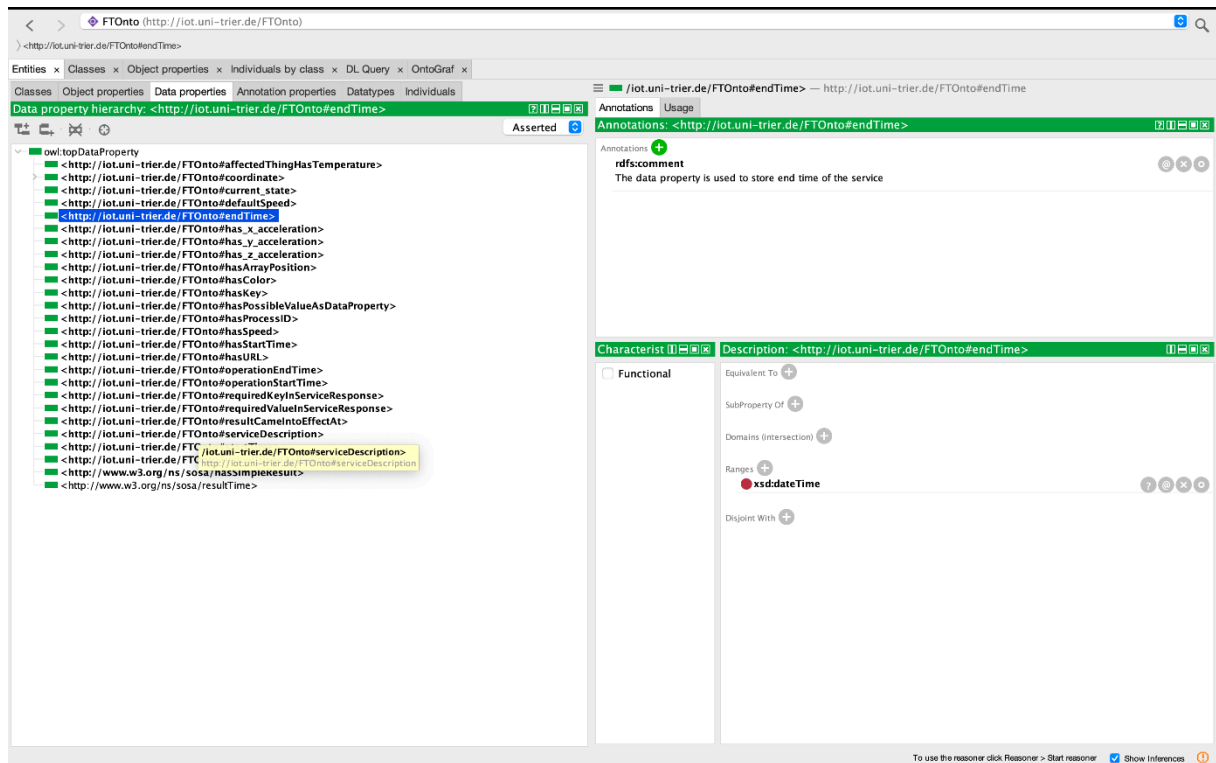
Each observation is linked to its associated execution instance through `has_service_execution`.

Changes in Ontology

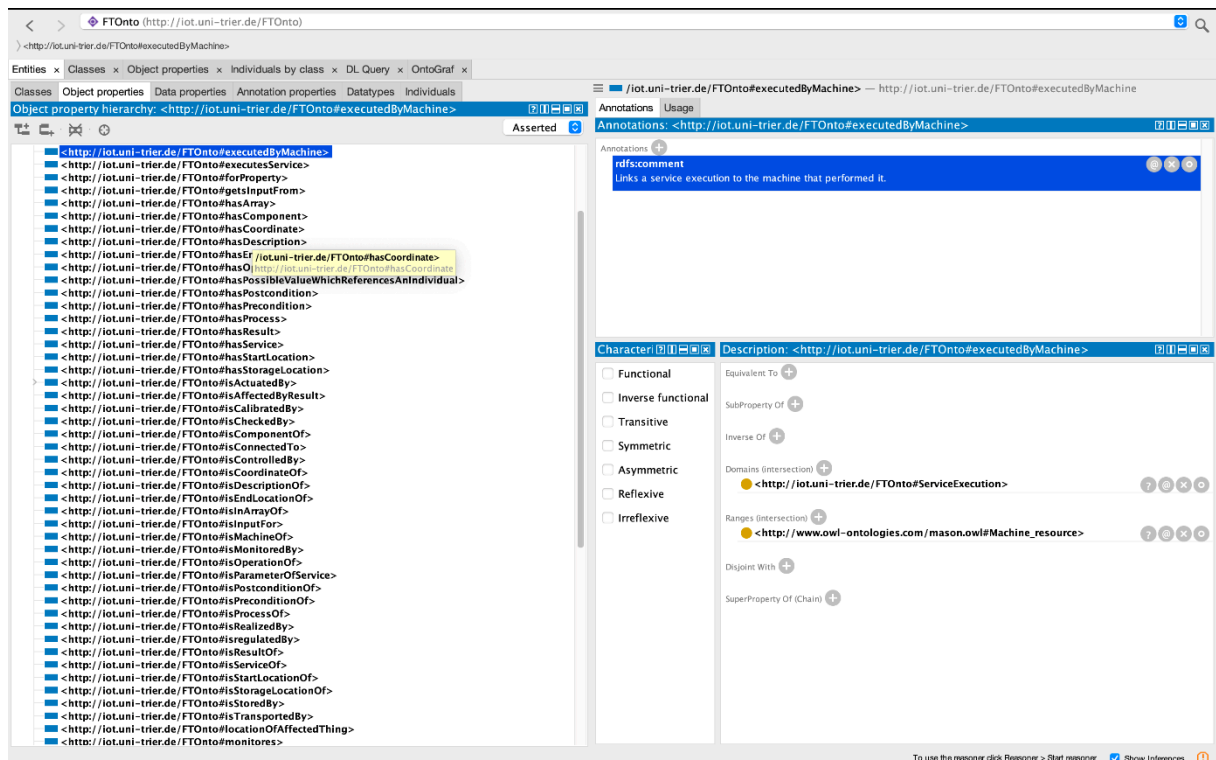
1. A data property, `startTime`, has been added to store the end time of service execution.



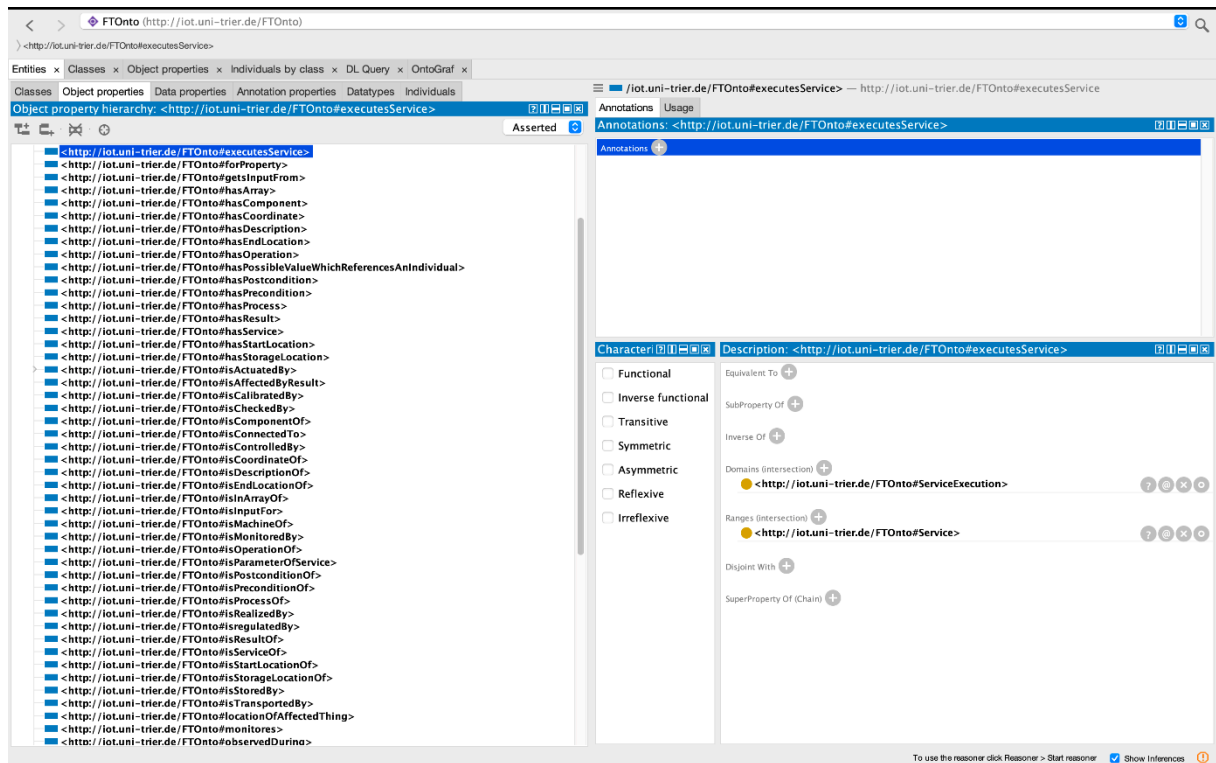
2. A data property, `endTime`, has been added to store the end time of service execution.



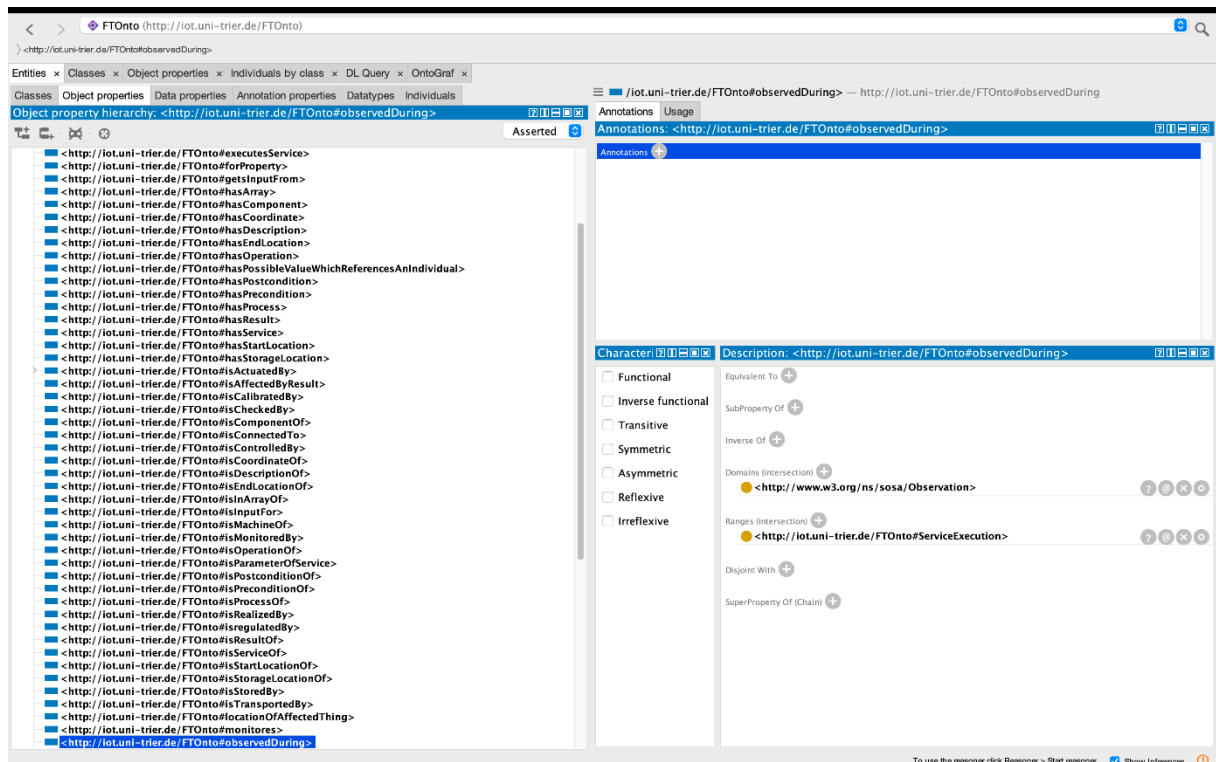
3. An object property, `executedByMachine`, has been added to link a service execution to the machine that performed it.



4. An object property, `executesService`, has been added to link a service execution instance to the abstract service it realizes



5. An object property, `observedDuring`, has been added to link service execution with observation



6.

Transformation Rules:

1. Transformation Rule for importing Pressure Observation data

```
{
  "Collections": [
    {
      "sheetName": "Tabelle1",
      "startColumn": "A",
      "endColumn": "G",
      "startRow": "2",
      "endRow": "+",
      "comment": "",
      "rule": "Individual: @A*\nTypes: \n\u0027Pressure Observation\u0027\nFacts: \u0027made by sensor\u0027 @C*, \n\u0027has simple result\u0027 @E*(xsd:float), \n\u0027result time\u0027 \r\n@F*(xsd:dateTime), \n\u0027observedDuring @G*",
      "active": true
    }
  ]
}
```

2. Transformation Rule for importing Acceleration Observation data

```
{
  "Collections": [
    {
      "sheetName": "Tabelle1",
      "startColumn": "A",
      "endColumn": "I",
      "startRow": "2",
      "endRow": "+",
      "comment": "",
      "rule": "Individual: @A*\nTypes: \u0027Acceleration Observation\u0027\nFacts: \u0027made by sensor\u0027 \r\n@C*, \n\u0027observedDuring\u0027 @I*, \n\u0027result time\u0027 \r\n@H*(xsd:dateTime), \n\u0027has_x_acceleration\u0027 \r\n@E*(xsd:float), \n\u0027has_y_acceleration\u0027 \r\n@F*(xsd:float), \n\u0027has_z_acceleration\u0027 @G*(xsd:float)",
      "active": true
    }
  ]
}
```

3. Transformation Rule for importing Barrier Observation data

```
{
  "Collections": [
    {
      "sheetName": "Tabelle1",
      "startColumn": "A",
      "endColumn": "G",
      "startRow": "2",
      "endRow": "+",
      "comment": "",
      "rule": "Individual: @A*\nTypes: \u0027Light Barrier Observation\u0027\nFacts: \u0027made by sensor\u0027 \r\n@C*, \n\u0027observedDuring\u0027 @G*, \n\u0027result time\u0027 \r\n@F*(xsd:dateTime), \n\u0027has simple result\u0027 @E*(xsd:float)",
      "active": true
    }
  ]
}
```